

- 194.** Let 1 man's daily earning be ₹ x and 1 boy's daily earning be ₹ y .

$$\text{Then, } 3x + 4y = \frac{756}{7} = 108 \quad \dots(i)$$

$$\text{And, } 11x + 13y = \frac{3008}{8} = 376 \quad \dots(ii)$$

Multiplying (i) by 11 and (ii) by 3, we get:

$$33x + 44y = 1188 \quad \dots(iii)$$

$$33x + 39y = 1128 \quad \dots(iv)$$

Subtracting (iv) from (iii), we get: $5y = 60$ or $y = 12$.

Putting $y = 12$ in (i), we get: $3x = 60$ or $x = 20$.

$\therefore$ Required time

$$= \frac{2480}{7x + 9y} = \frac{2480}{7 \times 20 + 9 \times 12} = \frac{2480}{248} = 10 \text{ days.}$$

- 195.** Let the price of 1 burger, 1 shake and 1 fries be ₹ x , ₹ y and ₹ z respectively. Then,

$$3x + 7y + z = 120 \quad \dots(i)$$

$$4x + 10y + z = 164.50 \quad \dots(ii)$$

Subtracting (i) from (ii), we get: $x + 3y = 44.50$

$$\text{or } 2x + 6y = 89 \quad \dots(iii)$$

Subtracting (iii) from (i), we get: $x + y + z = (120 - 89) = 31$.

$\therefore$ Cost of 1 burger, 1 shake and 1 fries = ₹ 31.

- 196.** $2x^2 + 12x + 18 = 0 \Leftrightarrow x^2 + 6x + 9 = 0$
 $\Leftrightarrow x^2 + 3x + 3x + 9 = 0$
 $\Leftrightarrow x(x + 3) + 3(x + 3) = 0$
 $\Leftrightarrow (x + 3)^2 = 0 \Leftrightarrow x = -3$.
- 197.** $x^2 - 7x = -12 \Leftrightarrow x^2 - 7x + 12 = 0$
 $\Leftrightarrow x^2 - 3x - 4x + 12 = 0$
 $\Leftrightarrow x(x - 3) - 4(x - 3) = 0$
 $\Leftrightarrow (x - 3)(x - 4) = 0$
 $\Leftrightarrow x = 3 \text{ or } x = 4$.

- 198.** $2x^2 - 11x + 15 = 0 \Leftrightarrow 2x^2 - 6x - 5x + 15 = 0$
 $\Leftrightarrow 2x(x - 3) - 5(x - 3) = 0$
 $\Leftrightarrow (x - 3)(2x - 5) = 0$
 $\Leftrightarrow x = 3 \text{ or } x = \frac{5}{2}$.

199. Given exp. = $\frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} \times \dots \times \frac{m-1}{m} = \frac{1}{m}$.

200. Given exp. = $\frac{4}{3} \times \frac{5}{4} \times \frac{6}{5} \times \dots \times \frac{n+1}{n} = \frac{n+1}{3}$.

201. Given exp. = $\frac{5}{3} \times \frac{7}{5} \times \frac{9}{7} \times \dots \times \frac{1001}{999} = \frac{1001}{3}$.

202. Given exp. = $\frac{3}{4} \times \frac{4}{3} \times \frac{5}{3} \times \frac{3}{5} \times \frac{13}{7} \times \frac{1}{13} = \frac{1}{7}$.

- 203.** Given exp.

$$= \left(\frac{3^2 - 1}{3^2} \right) \left(\frac{4^2 - 1}{4^2} \right) \left(\frac{5^2 - 1}{5^2} \right) \dots \left(\frac{11^2 - 1}{11^2} \right) \left(\frac{12^2 - 1}{12^2} \right)$$

$$= \left[\frac{(3+1)(3-1)}{(2+1)(4-1)} \right] \left[\frac{(4+1)(4-1)}{(3+1)(5-1)} \right] \left[\frac{(5+1)(5-1)}{(4+1)(6-1)} \right]$$

$$\dots \left[\frac{(11+1)(11-1)}{(10+1)(12-1)} \right] \left[\frac{(12+1)(12-1)}{(11+1)(13-1)} \right]$$

$$= \frac{(3-1)}{(2+1)} \times \frac{(12+1)}{(13-1)} = \frac{2}{3} \times \frac{13}{12} = \frac{13}{18}.$$

204. Given exp. = $\frac{1}{(2-1)(2+1)} + \frac{1}{(4-1)(4+1)}$
 $+ \frac{1}{(6-1)(6+1)} + \dots + \frac{1}{(20-1)(20+1)}$
 $= \frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{19 \times 21}$
 $= \frac{1}{2} \left(1 - \frac{1}{3} \right) + \frac{1}{2} \left(\frac{1}{3} - \frac{1}{5} \right) + \frac{1}{2} \left(\frac{1}{5} - \frac{1}{7} \right) + \dots + \frac{1}{2} \left(\frac{1}{19} - \frac{1}{21} \right)$
 $= \frac{1}{2} \left(1 - \frac{1}{21} \right) = \frac{1}{2} \times \frac{20}{21} = \frac{10}{21}.$

205. Given exp. = $\left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right)$
 $+ \left(\frac{1}{4} - \frac{1}{5} \right) + \dots + \left(\frac{1}{11} - \frac{1}{12} \right) = \left(1 - \frac{1}{12} \right) = \frac{11}{12}.$

- 206.** Clearly, sum of first 6 terms is zero. So, sum of first 30 terms = 0.

$$\therefore \text{ Required sum} = \left(\frac{1}{2} + \frac{1}{3} - \frac{1}{4} - \frac{1}{2} - \frac{1}{3} \right) = -\frac{1}{4}.$$

207. Given exp. = $(1 + 11 + 111 + 1111) + \left(\frac{1}{2} \times 4 \right)$
 $= 1234 + 2 = 1236.$

208. Given exp. = $\left(1000 - \frac{1}{1000} \right) \times 7 = \left(7000 - \frac{7}{1000} \right)$
 $= 6999 \frac{993}{1000}.$

209. Given exp. = $\left(1000 - \frac{4}{999} \right) \times 999 = 999000 - 4 = 998996.$

210. Given exp. = $\left(1000 - \frac{6}{7} \right) + \left(1000 - \frac{5}{7} \right) + \left(1000 - \frac{4}{7} \right)$
 $+ \left(1000 - \frac{3}{7} \right) + \left(1000 - \frac{2}{7} \right) + \left(1000 - \frac{1}{7} \right)$
 $= 6000 - \left(\frac{6}{7} + \frac{5}{7} + \frac{4}{7} + \frac{3}{7} + \frac{2}{7} + \frac{1}{7} \right)$
 $= 6000 - \frac{21}{7} = 6000 - 3 = 5997.$

Another Method:

$$\text{Given exp.} = (999 \times 6) + \left(\frac{1}{7} + \frac{2}{7} + \frac{3}{7} + \frac{4}{7} + \frac{5}{7} + \frac{6}{7} \right)$$

$$= 5994 + \frac{21}{7} = 5994 + 3 = 5997.$$

211. Given exp. = $(998 \times 5) + \left(\frac{2}{17} + \frac{3}{17} + \frac{5}{17} + \frac{8}{17} + \frac{16}{17} \right)$
 $= 4990 + \frac{34}{17} = 4990 + 2 = 4992.$

212. Given exp. = $\left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{9} - \frac{1}{10} \right)$
 $= \left(1 - \frac{1}{10} \right) = \frac{9}{10}.$